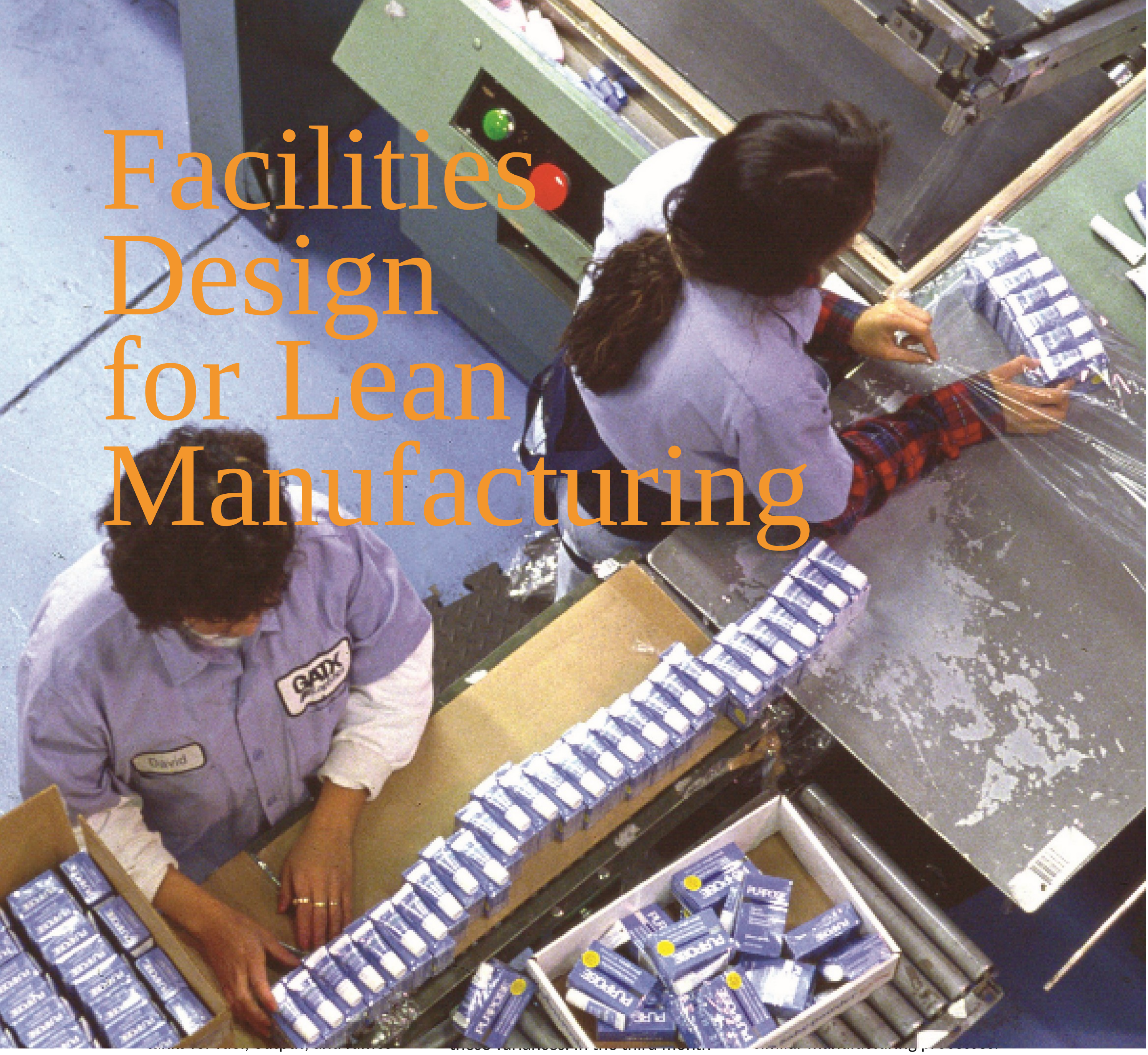

Facilities Design for Lean Manufacturing

BY KEVIN J. DUGGAN

(warehouse), we can make room for more engines and thereby generate more cash from our building.

In summary, smaller engines with variable output to match our daily customer demand each day would make our building generate more cash and grow our business.

the president calls and asks about the amount of shipping that is needed to make the quarter s numbers. The plant manager does everything in his power ... hires temporary unskilled labor, works overtime, etc., to get the products out. For some reason, the variance in the third quarter does

The next step is to group the products with similar patterns. An exact match of X patterns is not required to group products together. Products with 70 percent to 80 percent overlap can be made in the same cell. At this time, only a rough estimate is needed (Figure 2).

Pick a group of products to serve as

As manufacturers strive to become more competitive in the global marketplace, they look at emerging concepts such as lean thinking that might have the potential to focus their operations on significant bottom-line improvements. Lean thinking embraces concepts such as value-added services to reduce waste. Value specifications and value stream identification help manufacturers develop new products and bring them to market quicker with fewer resources. Flow and pull manufacturing com- prise the lean engine that produces these products to meet customer demand each day. A continuous

improvement program that would drive manufacturers toward perfection would encompass the complete lean cycle. Many opportunities exist to improve operations through the use of lean concepts. However, the best opportunity may lie in a facilities design project. If we had the opportunity to design a new factory or redesign an existing facility using lean concepts, what tools would we use? Would the traditional tools of facility design be useful in applying the concepts of lean manu- facturing? Can a department relation- ship chart be used as a tool to design a factory with no departments? Can

algorithms that optimize traffic be effective in a factory warehouse? Obviously, the warehouse with no traditional tools do not apply to designing the lean factory. In fact, a new set of tools is needed.

An analogy

put at a fixed rate, the building would

Processes													
Products	Casting	Polishing	Threading	Assembly	Bend	Drill	Wash	Plating	Welding	Paint	Stamping	Riveting	Final Pack
787	X			X		X	X	X		X	X	X	X
4210	X	X	X	X									X
909	X			X	X	X	X	X	X	X	X		X
6211	X	X	X	X									X
2528	X		X	X		X	X		X	X	X		X
3798						X	X			X	X	X	X
5736	X		X	X									X
806	X			X	X		X	X	X	X	X		X

Figure 1. A process map reveals patterns that show which products follow similar manufacturing processes.

Processes													
Products	Casting	Polishing	Threading	Assembly	Bend	Drill	Wash	Plating	Welding	Paint	Stamping	Riveting	Final Pack
4210	X	X	X	X									X
5736	X		X	X									X
6211	X	X	X	X									X
2528	X		X	X		X	X		X	X	X		X
3798						X	X			X	X	X	X
909	X			X	X	X	X	X	X	X	X		X
787	X			X		X	X	X		X	X	X	X
806	X			X	X		X	X	X	X	X		X

Figure 2. Group the products with similar patterns to determine the number of manufacturing cells required.

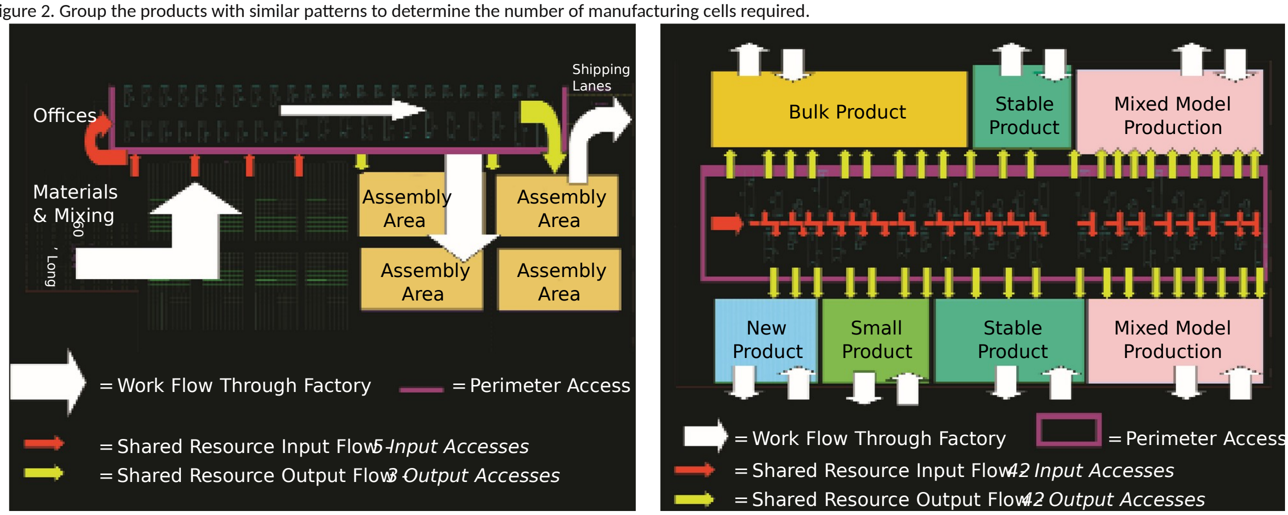

Figure 3. A traditional facility flow map (left) reveals little perimeter access, while a lean facility map (right) shows the flexibility afforded by process-based cells.

a pilot project and perform resource calculations based on the cycle times, yields, and volumes for each product through its respective processes. (Remember that resources need to be rounded up: If the number of resources equals 2.2, three are required.)

After working with this spreadsheet you would begin to develop the total number of cells required and the resources needed for each cell. Next, look at space requirements. Add a column to the spreadsheet that indicates the space required for each workstation. Multiply that by the number of stations needed, and you have a rough idea of the cell size. At the end of this step, you have the number of different cells required for you're factory and the size needed for each cell. Now you are ready for our lean layout.

Shared resources In most cases, the cells we developed cannot run completely independently. Most likely they will need to pull from a shared resource (a process or piece of equipment that supplies multiple cells with parts). These shared resources need to be set up to run the various parts needed to feed different cells. In order to flow product on demand, enough inventory is needed to cover the set-up time of the machine and replenish the cell when it calls for parts. This buffer inventory can be stored in or next to the cell. The quicker the machines can be set up, the less inventory is needed to buffer. Programs such as the SMED (singleminute exchange of die) are very effective in reducing set-up times.

Locating shared resources in the factory is a crucial step in the flow design. Typically, machine banks (such as injection molding or stamping machines) are placed along a wall that houses the utilities needed to run them. This makes sense for machine installation and may yield a one-time reduction for installation. However, perimeter access is very important to achieving value-added facilities design. Ideally, parts would be flowed out of machines directly through the remaining processes until they are shipped to customers. With a shared resource, there should be direct flow to as many cells as possible. Placing the shared resource along a wall restricts the flow pattern and allows material to flow one way only. Hence, material movement (which is a non-value-added activity) is increased. Placing a shared resource in the center of the factory where it can feed as many cells as possible provides the least material movement as well as more visibility to the shared resource.

A way to view the resource flow is through the use of a shared resource flow map. Show the resource flows on your drawings. While distance traveled and intensity are important to know, more significant are the amount of flows available to feed adjacent cells directly. The drawing that has the most adjacent cells with direct flows is the preferred option. Also keep in mind that material should be fed to the shared resource by using the height of the building whenever possible. Feeding pellets into a hopper can be done through vacuum pumps and piping in the ceiling, eliminating the need to occupy floor space with rawmaterial buffers. Resource flow maps should identify the number of flow paths going into and out of the shared resource. Remember that the product line will change in the future and cells will need to be reconfigured. The number of adjacent cell feeds and the flow paths through the shared resource will determine the adaptability of this cash-generating machine to meet future needs.

Cell types

The types of cells in the factory are not based on the products they produce but the processes needed to produce a group of products (Figure 3). From an operational standpoint, different cell types are used to match demand. For example, a cell type that produces stable items (which may be considered the company s basic product that has been produced for years) should be considered. The advantage of this *stable product* type of cell is that its required support resources (e.g., production control, engineering) are very low; therefore, resources can be used in other cells that need them more (such as those that produce new products). Other types of cells to consider are:

Bulk product cells The size of the product is large and requires space to be directly produced and shipped. This type of cell should have shipping and receiving door access within the cell.

Small product cells The size of the product is small and requires less space than a normal product.

New product cells Cells that focus on the launch and debugging of new products. At a later point these can be moved into another cell type as demand changes.

Mixed model cells For products that are no longer at capacity. Multiple products can be run in the same cell each day in order to match daily demand.

Designing the adaptable value stream

For a facility to be lean, the value stream for its products must be understood. Continuous improvement and changes to match the value stream to the current product line will be necessary. But the facility has to be designed to accommodate change. What design elements are needed to accommodate this?

*Work stations*Modular workstations can be rearranged quickly to make new cell configurations. However, workstations need power, compressed air, and lighting. In many cases, these are supplied by utilities in the ceiling. To keep flexibility in workstation design, independent mobility must be part of the plan. A workstation should have its own lighting, but it should be powered with electricity and compressed air without going to the ceiling. One method for doing this is to supply each workstation with male/female connectors that can be daisy-chained together. One workstation powered from an existing drop in the ceiling could power the workstation next to it through electrical cords and rubber air lines. This station would then power the next station in the same manner, and so on.

*Shared resources*Independent workstations build flexibility into work cells, but what about the shared resources? Say there is a bank of machines designed to feed cells adjacent to it. What happens when the business changes and machine A at one end of the factory has to feed cell Z at the other end? In fact, a year from now the flow patterns can change completely, causing more non-value-added activities than before.

A concept of parking spaces for machinery can help. If each

A new language for facilities design

Adaptability: The ability to change the shipped. This includes order administration, factory configuration quickly to match processing, planning functions, work-order new product lines or changes in the pro- generation, manufacturing, and many other duction mix.
majority of which are non-

Capacity: The maximum amount of value-added. This area is a prime target for demand a factory can meet over a certain lean manufacturing since identifying the amount of time. Factory capacity is never value stream of our products here will really fixed because product mix and greatly reduce non-value-added activities.

product lines can change. A reduction in *Scope*: The scope of a lean facility
warehouse space can make room for design includes high throughput speeds,
more manufacturing space, which will various types of cells to match
necessary increase capacity. products and processes, flexibility to match

Capacity is also dependent upon mak- varying demand, the performance
of valueing the right product at the right time. If added activities only,
adaptability to future you are manufacturing in batches, you may product
lines, and an ability to continually be locked in to a specific run on a specific
increase capacity based on improvements. product and not be able to produce
the *Throughput*: The time from the start of product you need that day.
In fact, youthe first process to the completion of the may send jobs out to a
vendor while you last process. This is measured from the have the capacity
in-house. Running the time the first machine for the first process wrong product
at the wrong time reduces is set up until the first good finished piece
capacity. Reducing throughput times on is completed. The smaller the
batch sizes, products will increase capacity. the quicker the throughput speed.

Flexibility: The ability to vary the output Manufacturing to a lot size of one
yields of production to meet daily demand. optimum throughput speed.
Flexibility is sacrificed if a factory is locked *Value-added services*: Anything the cusinto a
specific rate of production in order tomer is willing to pay for in a product. to
utilize equipment or make a line effi- Customers do not care whether you have a
cient. The more flexible a factory is, the large warehouse and stock thousands of
more its capacity increases. items for immediate delivery or if you make

Manufacturing cell: A group of manu- items to order with no inventory, as
long as facturing operations located together that product is available when
they need it. contain all the processes needed to take a Manufacturers that
can build to order withproduct from raw material to finished good. out
inventory are at an advantage since *Perimeter access*: The amount of expo-
their pricing will be more competitive. sure at the borders of a shared
resource. *Volatility*: The variability of demand for This is particularly important
for flow and a product. A highly volatile product is pull manufacturing because
various cells ordered sporadically with wide swings in pull work through a
common process. the quantity ordered each time. A stable Accessibility to this
process is crucial. item is one whose demand is consistent. A

Response time: The time from the point volatile product consumes more
resources; at which an order is received until it is a stable product
consumes less.

machine in a machine bank were equipped with a temporary parking space that could accommodate any machine in the bank, then any machine could be placed anywhere within the bank. For example, a parking space for a 125-ton injection-molding machine would have the same power needs and footprint size as a 500-ton machine. So two 125-ton machines could be put in one parking space if the cell configuration warranted.

The next step is to design in quick disconnects for machines. Air lines, electrical power, and waste removal systems should be detachable and reattachable to any machine in the machine bank. Built-in guides for forklifts should be placed in the frame of the machine. Similar to a set-up reduction program, a machine relocation program should be in effect to move equipment as quickly as possible.

*The value stream*Performing the cellular designs and resource flow maps mentioned before will yield an effective design for a given point of time. Non-value-added activities are then greatly reduced, and value streams are developed for the products that the cells accommodate. Designing in the flexibility factors will yield the ability to redesign the facility very quickly to match product and demand changes, which will effectively increase capacity. With this factory design, the cellular development and resource flow maps could be redrawn every quarter or more. The factory could then be adjusted easily by moving the resource equipment into the correct positions to minimize non-value-added activities and create value streams for the manufacturing of future products.

There are many obvious benefits to creating flexible cells that can produce product from raw material to finished goods. The drastic reduction of material handling, inventory, and manufacturing lead-time will flush substantial dollars to the bottom line. However, the operational benefits to a lean facility design can yield more important benefits, those that impact future growth. By using the tools previously mentioned for lean facilities design, we have poured the foundation for simple operational systems that can support it. For example, to determine if there is enough inventory on hand to produce a variety of products, a quick look into the cell can provide the answer. This eliminates the non-value-added steps of checking a computer system for each part, then trying to determine if the computer report is accurate or not.

Still more powerful is the ability to link sales and marketing directly to the manufacturing cells that produce their product line. In a conventional environment, checking the status of an order would follow a scenario similar to this:

The sales representative calls the coordinator, who calls the corporate operations planning manager, who calls the corporate planner, who calls the plant production control manager, who calls a meeting with the planners to check the status of all the material needed to manufacture the product. Each planner then checks with each production supervisor to provide estimated completion dates. This information funnels back to the local production control manager, then to the corporate product planner, then to the operations planning manager, then back to the sales coordinator, then back to the salesperson. This entire activity is non-valueadded and could take days or weeks to complete.

In a lean facilities design, if the status of an order has to be checked, a direct call from the sales representative or customer can be made directly to the production cell team leader. The team leader would then perform a quick visual survey of the current production cell and provide the answer. This two-step approach greatly reduces non-value-added activities. And it s possible that checking the status of orders might never need to happen if the manufacturing process is so streamlined that orders are turned around in 24 to 48 hours. The more non-value-added activities eliminated, the more competitive a company becomes, and the more the business grows.

As many companies embrace lean manufacturing concepts, facilities design will provide a foundation that may determine the fatness of the structure. A facilities design that considers flexibility and adaptability will allow for a structure that can be redesigned time and again.

For further reading

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